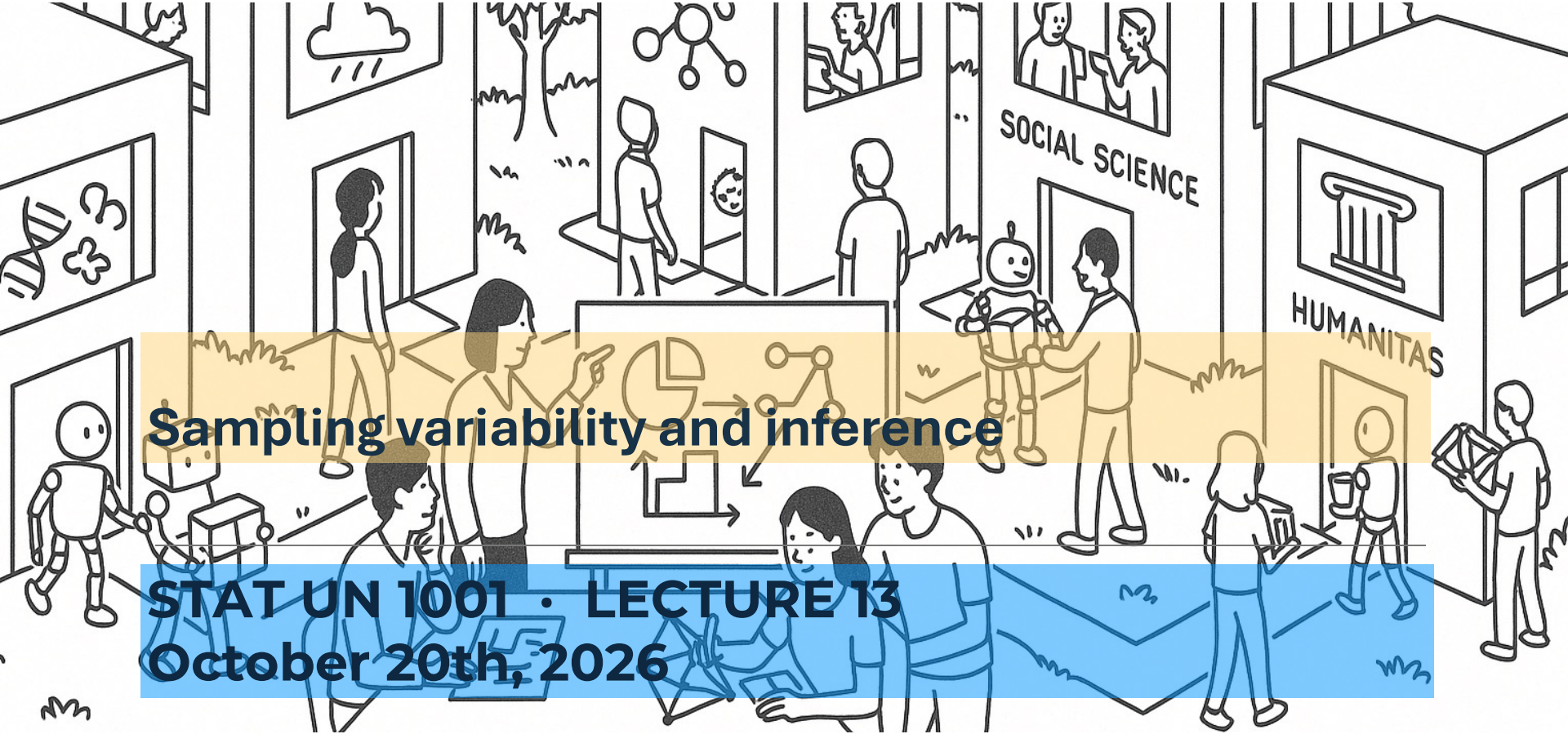


Sampling variability and inference

STAT UN 1001 · LECTURE 13
October 20th, 2026

Week 7 response map

55 check-ins

25 practice-quiz submissions

First review priority	Responses	Share
Sampling distributions of means	19 / 55	35%
Binomial counts	16 / 55	29%
Normal tails / intervals	11 / 55	20%
Model assumptions	9 / 55	16%

The hardest practice items were normal intervals (60%) and estimated standard error (56%). We will work those slowly, then connect them to the larger inference story.

Reasoning moves from models to inference

Question	Model language	Inference language
What can happen?	Outcomes and events	Information in a Sample about the Population
How often?	Probability	Sampling variability
What number do we report?	Area or formula	Statistic and point estimate
How much trust?	Check model fit	Check bias and standard error

Workflow: choose the model, name the event or parameter, calculate only after the setup is clear, and finish with a sentence in context.

Normal-model intervals

A probability sentence comes before the calculator

Step	What we write	Why
1	State the event: $60 < X < 65$	Name the values we care about
2	Convert both endpoints to z	Put the problem on the standard scale
3	Subtract $P(Z < z_{low})$ from $P(Z < z_{high})$	Keep only the area between the cutoffs
4	Write the result in context	A decimal needs a meaning

Exercise 4.44: heights of 10-year-olds

Use X ~ normal with mean 55 inches and standard deviation 6 inches.

Let’s set up each question before calculating.

Part	Question	Setup to show
a	What proportion is shorter than 48 inches?	One left-tail z-score
b	What proportion is between 60 and 65 inches?	Two z-scores and an area difference
c	What height marks the tallest 10%?	The 90th percentile, transformed back to inches

Appendix C.1: read the normal table

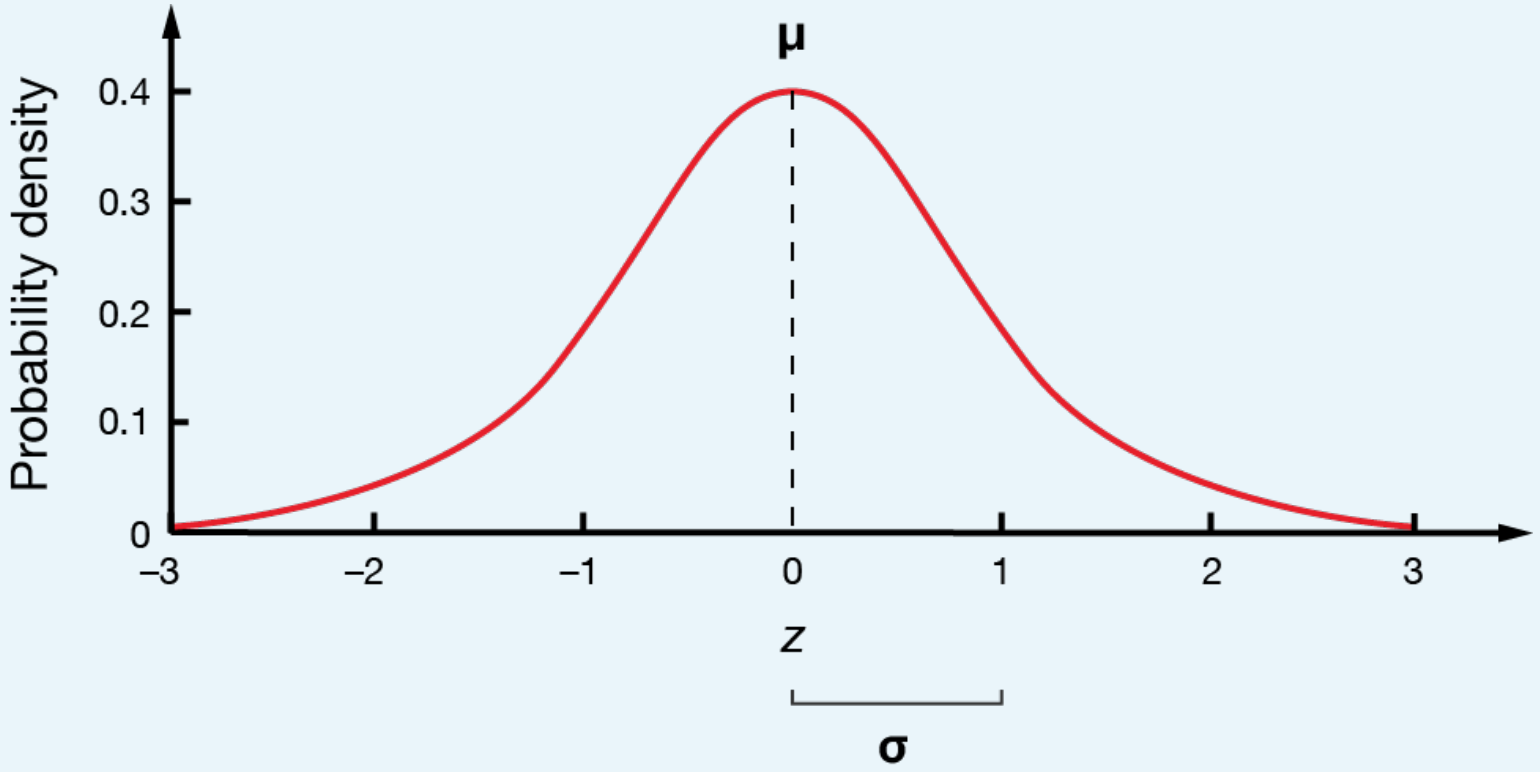
The table reports the area to the left of a Z-score

	Second Decimal Place				
Z row	**0.00**	**0.01**	**0.04**	**0.05**	**0.09**
0.4	0.6554	0.6591	0.6700	0.6736	0.6879
1.0	0.8413	0.8438	0.8508	0.8531	0.8621

Read the row through the first decimal place, then use the column for the second decimal place. Example: z = 0.45 uses row 0.4 and column 0.05, giving 0.6736.

Appendix C.1 gives an abbreviated table on p. 408 and the full table on p. 410.

Appendix C.1: read the normal table



Appendix C.1: read the normal table

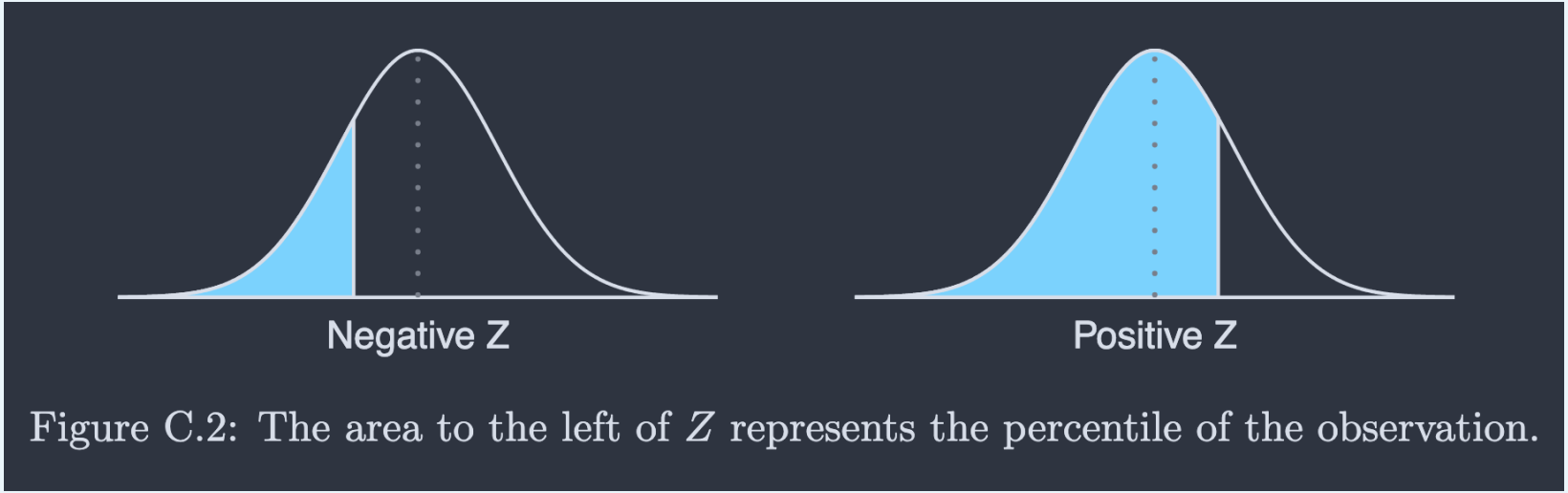


Figure C.2: The area to the left of Z represents the percentile of the observation.

Example C.1: Ann's SAT percentile

SAT scores: X ~ normal with mean 1100 and SD 200
Ann's score: x = 1300

Step	Work	Meaning
1	z = (1300 - 1100) / 200 = 1.00	Ann is 1 SD above the mean
2	Row 1.0, column 0.00 -> 0.8413	Area to the left of z = 1.00
3	0.8413 = 84.13%	Ann is about the 84th percentile

Percentile means the percentage of observations below the score.

Example C.2: upper tails and reverse lookup

Question	Table move	Answer
P(Z > 1.00)	1 - P(Z < 1.00) = 1 - 0.8413	0.1587, or 15.87%
Z for the 80th percentile	Find the table entry closest to 0.8000	0.7995 ➡️ z = 0.84

The normal table gives left-tail area.
For a right tail, subtract from 1.
For a percentile-to-Z question, search the table body first.

Notice the direction: a high score has a large left-tail percentile but a small right-tail area.

Example C.2: upper tails and reverse lookup

	Second decimal place of Z									
Z	*0.00*	0.01	0.02	0.03	**0.04**	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	**0.7995**	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	*0.8413*	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮

Figure C.1: A section of the normal probability table. The percentile for a normal random variable with $Z = 1.00$ has been *highlighted*, and the percentile closest to 0.8000 has also been **highlighted**.

Notice the direction: a high score has a large left-tail percentile but a small right-tail area.

Try it: use the table

Name the row, column, and tail before finding the area.

Prompt	Answer based on Table entry
1. Find $P(Z < 0.33)$	Row 0.3, column 0.03
2. Find $P(Z > 1.15)$	$1 - P(Z < 1.15)$
3. Find z at the 70th percentile	Search for 0.7000; use the closest entry

Exercise 4.44: between 60 and 65

Set up the calculation together

z_low = (60 - 55) / 6 = 0.83
z_high = (65 - 55) / 6 = 1.67

$$P(60 < X < 65) = P(0.83 < Z < 1.67) = P(Z < 1.67) - P(Z < 0.83)$$

Now calculate the area. Then write: "About ___% of 10-year-olds are between 60 and 65 inches, under this model."

Check: can you draw a standard normal curve and mark the areas corresponding to the table entries and explain the calculation?

Binomial event language

Exercise 4.24 uses the same chance-process discipline

Wording	What it asks	Typical setup
Exactly two	A count	P(X = 2) with choose(3,2)
At least one	A count plus its complement	1 - P(X = 0)
The third child is the first diseased child	One ordered path	not diseased, not diseased, diseased

Before any arithmetic: identify one trial, the success, n, p, and whether order matters.

Exercise 4.24: choose the setup

Three children. Each has probability p = 0.25 of having the disease. Treat the trials as independent.

Question	Choose one setup
Exactly two children have the disease	P(X = 2) = choose(3,2)(0.25)^2(0.75)^1
None have the disease	P(X = 0) = (0.75)^3
At least one is neither disease nor carrier	Use a complement
The first diseased child is the third child	(0.75)(0.75)(0.25)

Why the first and fourth rows are not the same kind of question?

Exercise 4.40: one bulb versus a sample mean

	One bulb	Mean of 15 bulbs
Center	mu = 9,000	mu_xbar = 9,000
Spread	sigma = 1,000	sigma_xbar = 1,000 / sqrt(15)
Cutoff	x = 10,500	xbar = 10,500
Standardize	(10,500 - 9,000) / 1,000	(10,500 - 9,000) / (1,000/sqrt(15))

Averages are less variable than individual observations. The center stays the same.

A sampling distribution describes repeated samples

Imagine repeating the same random sampling process many times.

One sample	Many repeated samples
One statistic, such as p-hat = 0.127	A distribution of possible p-hat values
One estimate of the parameter	A way to describe typical sampling variation
May be above or below the parameter	Centered near the parameter for a sound design

Sampling variability is not a mistake. It is the variation we expect when a chance process selects a different sample.

Inference vocabulary

Object	Meaning in the chip story	Symbol
Population	The larger production process of interest	all chips
Parameter	The unknown long-run severe-defect proportion	p
Sample	The 212 inspected chips	n = 212
Statistic	The defect proportion computed from the sample	p-hat
Point estimate	The statistic used as one estimate of p	27/212 = 0.127

A parameter belongs to the population. A statistic belongs to the sample.

Exercise 5.1: identify the parameter

For each situation, decide whether the population parameter is a mean or a proportion.

Ask first	Mean	Proportion
What kind of response does each person or case have?	A numerical measurement, such as hours or a score	A yes/no or category event, such as defective/not defective
What does the parameter summarize?	The population average	The population fraction in the event

A reported percentage can still be a mean if each person's response is numerical. Look at the response, not the formatting.

Exercise 5.3: estimate and uncertainty

212 chips inspected
27 severe defects

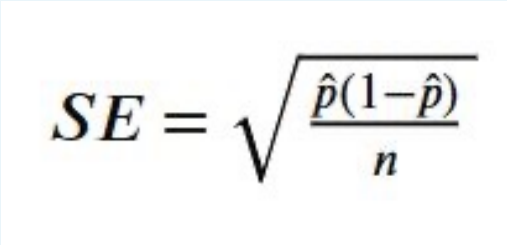
Historical rate: 10%

Point estimate

p-hat = 27 / 212 = 0.127

Estimated standard error

SE(p-hat) = sqrt[p-hat(1-p-hat)/n]



What does 0.127 estimate?
What does it not prove?
Why do we report an uncertainty measure too?

Precision and bias are different

	Small sample	Large sample
Sound sampling process	More variable estimates	More stable estimates
Biased sampling process	Variable and systematically shifted	More stable, still systematically shifted

A larger sample can improve precision. It cannot repair undercoverage, self-selection, or a flawed measurement process.